ESP32 SMART HOME

System Design and Demonstration

Project Team

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Presentation Outline

01 System Design

Users, requirements, architecture, budget and team roles

02 Hardware Demo

03 App Demo

1. Users and the Problem

Intended users

Residents who want to control a room's light and model door from a phone. The team demonstrates this use case with a small smart-home model on a shared LAN.

Need and expected benefit

- Control devices without going to each switch.
- See motion and device activity on the interface.
- Reduce unnecessary lighting through inactivity rules.

User profile is a working assumption. Energy savings remain unmeasured.

1. Scope and Functional Requirements

ID	Required behavior
FR1	The phone turns the light ON and OFF.
FR2	The phone opens and closes the model door.
FR3	PIR detection turns the light on, followed by off after a configurable inactivity period.
FR4	RFID checks an allowlist and opens the door for an authorized card.
FR5	The app displays light, door, motion and connection state.
FR6	The app shows recent sensor, access and device events.

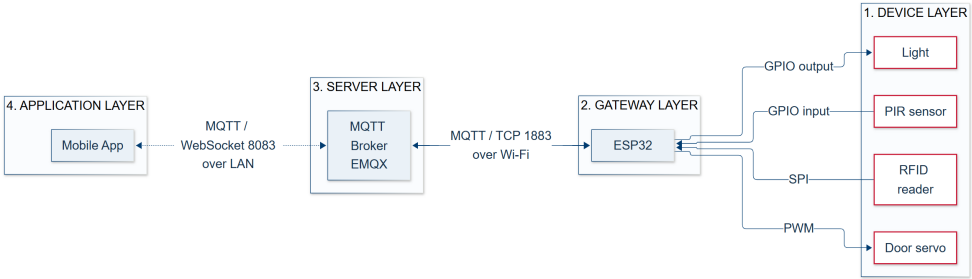
These are acceptance requirements from the task sheet. Voice is planned after core acceptance.

1. Non-functional Acceptance Targets

ID	Quality requirement	Acceptance criterion
NFR1	Operational stability	Run for 30 minutes without unintended reset or connection loss.
NFR2	Mechanical reliability	Complete 20 door cycles without jamming.
NFR3	Recovery and consistency	Reconnect after a network/broker interruption and restore current device states.
NFR4	Input robustness	Reject and log invalid control payloads while keeping firmware running.
NFR5	Reproducibility	All Must acceptance scenarios pass twice consecutively.

Targets come from M3-11, M3-08, M1-03/04/10, M1-06 and M5-09. Measurement results are pending.

1. Four-layer Architecture



The original components form four logical layers. ESP32 acts as the gateway. The mobile connection needs MQTT contract alignment and WebSocket configuration.

1. Technology Choices and Rationale

Technology	Rationale for this prototype
ESP32 / PlatformIO	Wi-Fi and device interfaces in a shared firmware project.
MQTT / EMQX	Pub/sub messaging for commands, states and events.
React Native / Expo	Shared mobile screens with MQTT connectivity.
Python + Paho	Test commands and display incoming device logs.
Docker Compose	Repeatable broker and service setup on a laptop.
Log storage	Server RAM holds 300 logs. App storage retains 50 logs and settings.

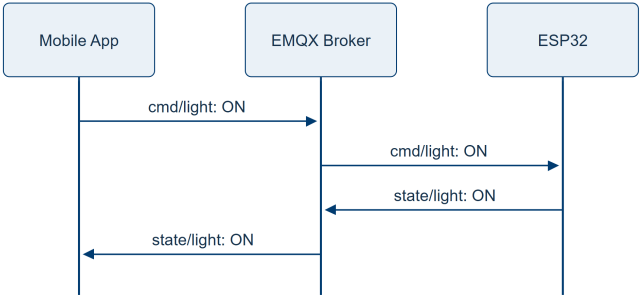
Core scope uses session/local logs. Persistent server storage remains a future design decision.

1. Bill of Materials

Component / set	Availability / budget (VND)
ESP32 WT-01 (Ethernet and Wi-Fi)	Available
USB–TTL CH340 adapter	Available
MB-102 breadboard	Available
SG90 servo	46,116
RC522 RFID module	Available
5V relay	Available
HC-SR501 PIR sensor	34,560
LM2596 buck module	45,144
Jumper wires (three types)	Available
12V 2A adapter	46,116
DC power jack	11,124
Budget for remaining items	**183,060**

Available items are already owned. RC522 selection remains to be confirmed.

1. Light Command and State Feedback



Original contract with topic prefix `smarthome/`. Alignment with the current JSON schema is pending.

1. Current Sensor-to-interface Flow



The hardware branch contains this path in code. The server stores recent logs in RAM and serves the web dashboard.
Durable database storage is a pending scope decision. Physical sensor-to-screen evidence still needs to be recorded.

1. MQTT Contract Alignment

	Hardware / test server	Current mobile code
Command topic	`smarthome/cmd`	`smarthome/cmd/light` `smarthome/cmd/door`
Feedback topics	`smarthome/logs`	`smarthome/state/+` `smarthome/event/access`
Payload	JSON command and log schema	Text commands and state values, JSON access events
Broker transport	MQTT over TCP 1883	MQTT over WebSocket 8083
Integration priority	Agree one version, update both clients and rerun shared test cases.	

1. Team Roles and Shared Repository

Member	Name	Student ID	Primary ownership
M1	Trần Sỹ Nguyên	20235985	Wi-Fi/MQTT core and firmware interfaces
M2	Lăng Hồng Nguyệt Anh	20235885	PIR, RFID authorization and automation
M3	Luong Ngoc Vu Long	20235967	Power, wiring, actuators and physical model
M4	Ha Duc Duong	20235923	Mobile dashboard, MQTT control and event UI
M5	Nguyen Vu Anh Khoa	20235957	Broker, integration, QA and report coordination

github.com/itstheblank/esp32_smart_home

Hardware Demo

Add the model photo, wiring view or live demonstration here.

App Demo

Add phone screenshots, a recording or the live app here.

Thank You!

Questions and Discussion